

Global CO₂ Emissions

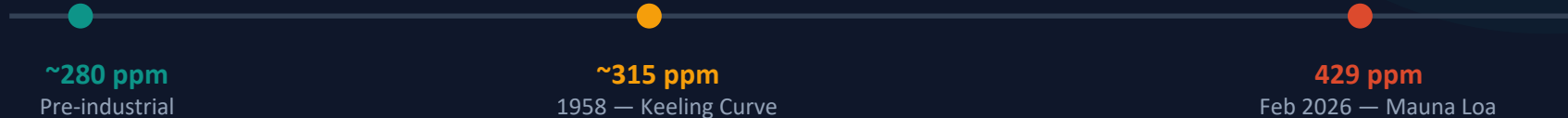
Trends, Regional Shifts, and Per Capita Inequalities

— **1750 – 2024**

Atmospheric CO₂ Reaches 429 ppm

50% above pre-industrial levels (~280 ppm)

429
ppm

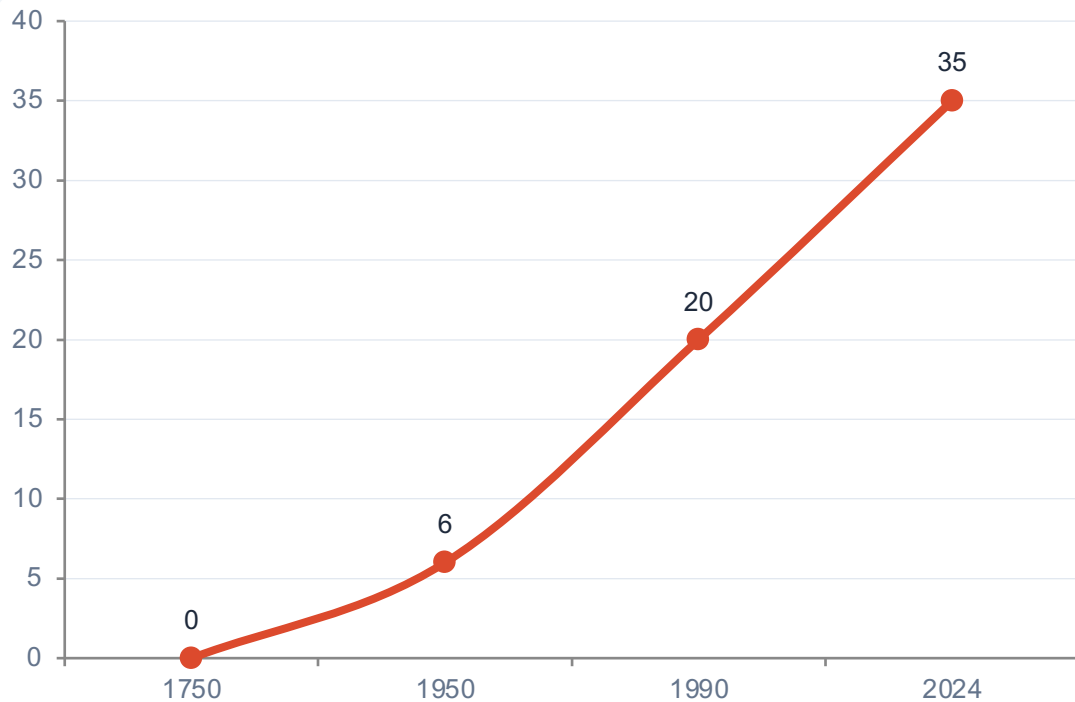


Current CO₂ levels are at least 1.5× the natural increase at the end of the last ice age — a change that took thousands of years then, but roughly two centuries now.

Source: NOAA Mauna Loa Observatory, February 2026



Global Fossil Fuel CO₂ Emissions Grew Six-Fold



1950

~6 Billion Tonnes

Post-war industrial expansion

1990

>20 Billion Tonnes

Nearly 4× since 1950

2024

35+ Billion Tonnes

Not yet peaked; growth has slowed



China Now Emits Over 12 Billion Tonnes

Country	Annual CO ₂	Share	Trend
China	~12.5 Bt	~27%	<i>Slight increase</i>
United States	~5.0 Bt	~14%	<i>Declining</i>
India	~3.0 Bt	~8%	<i>Rising</i>
EU-27	~2.7 Bt	~7%	<i>Declining</i>
Germany	~0.6 Bt	~1.7%	<i>Declining</i>
United Kingdom	~0.3 Bt	~0.9%	<i>Declining</i>
Brazil	~0.5 Bt	~1.4%	<i>Stable</i>
France	~0.3 Bt	~0.8%	<i>Declining</i>

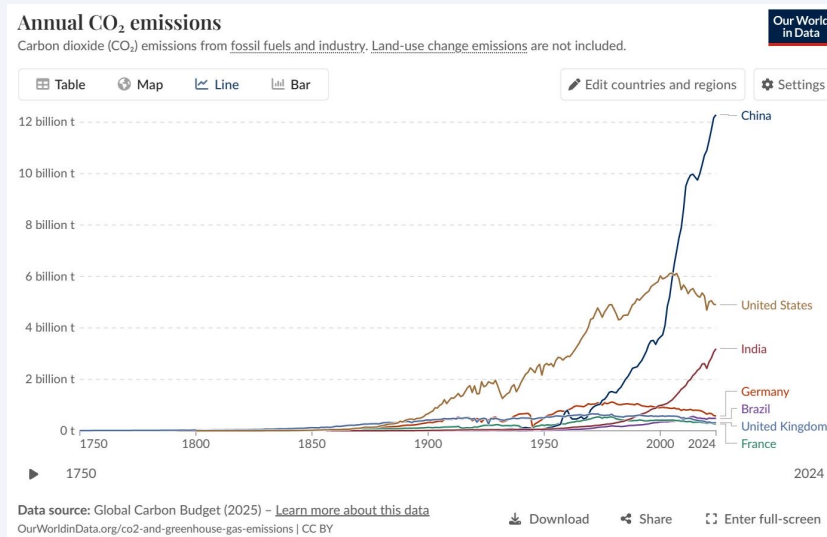


Figure: Annual CO₂ emissions by country

China emits more than double the United States and over one-quarter of the global total. US emissions have declined from a peak near 6 Bt around 2005.



Asia Now Accounts for ~50% of Global Emissions

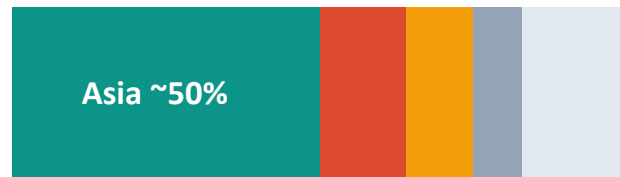
1900



Europe and the United States dominated global emissions, accounting for more than 90% of the total. By 1950, they still represented over 85%.



2024

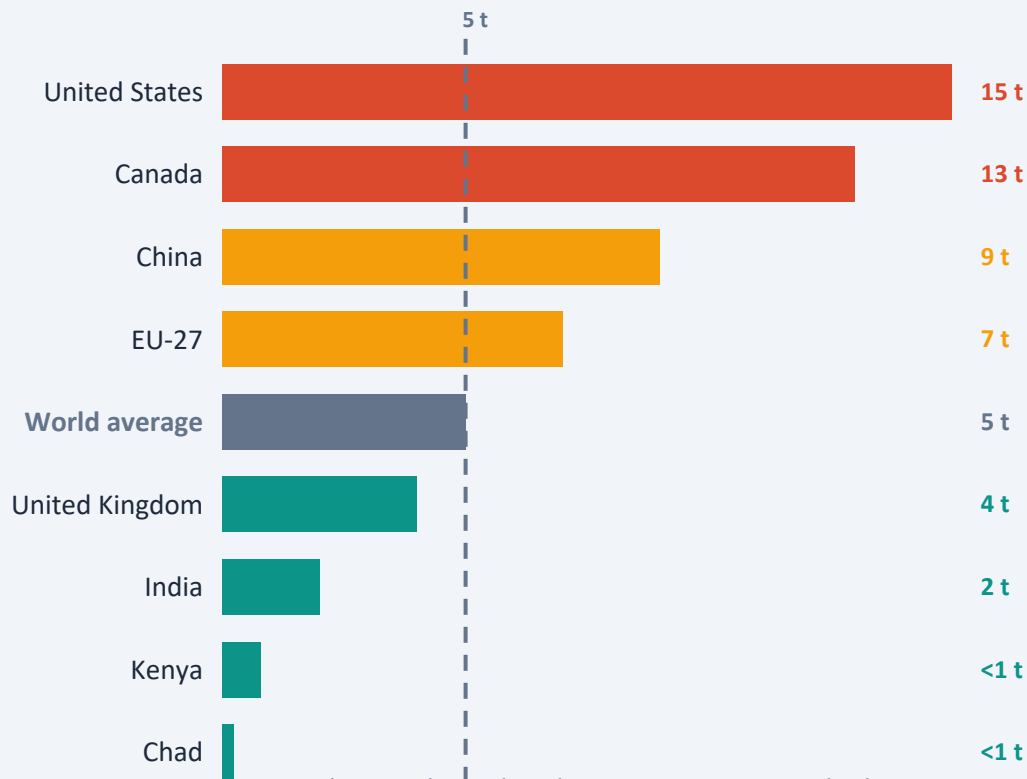


■ Asia ~50% ■ US ~14% ■ Europe ~11% ■ Africa + S. America ~3-4%

US + Europe combined now account for less than one-third of global emissions. Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.



Per Capita Gap: 15 t in the US vs. 0.1 t in Chad

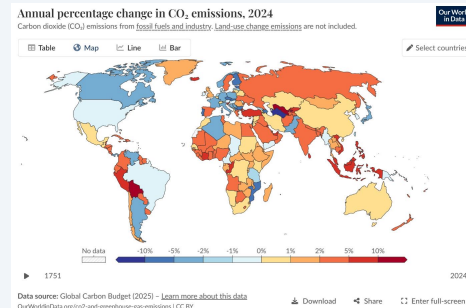


The average American emits in under two days what the average person in Chad emits in an entire year.

Source: Global Carbon Budget 2025; Our World in Data

150×

disparity between highest
and lowest per capita emitters



Per capita CO₂ over time



2024 Snapshot: US and Europe Cut Emissions

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table

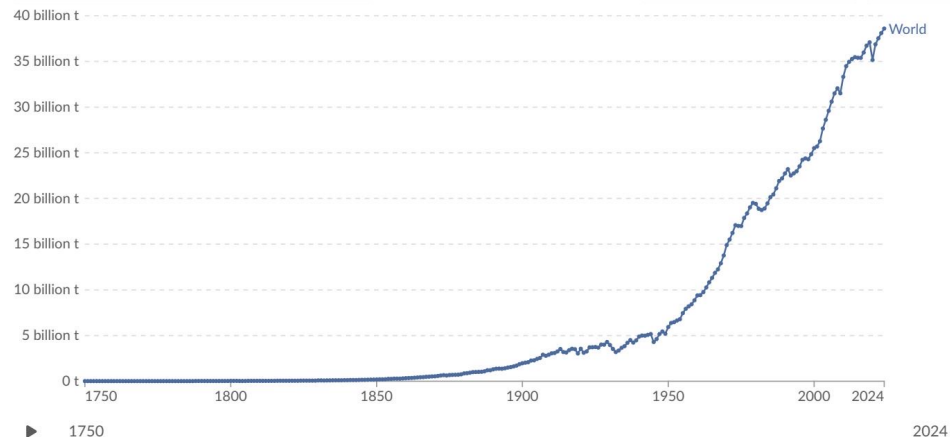
Map

Line

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Edit countries and regions

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Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Annual percentage change in CO₂ emissions, 2024

Declining (Blue)

- United States
- Much of Europe

Roughly -1% to -5% annual change

Increasing (Orange/Red)

- Russia
- Southeast Asia
- Parts of Africa

+2% to +10% or more annual change

Energy Mix Explains Why France Emits Far Less Than Germany

Countries with similar living standards can have very different per capita emissions depending on energy sources.

Lower Per Capita — ~4 t/yr

	France	UK
Nuclear	High	Moderate
Renewables	Growing	Growing
Fossil Fuels	Lower	Lower
Per Capita CO ₂	~4 t	~4 t

Higher Per Capita

	Germany	Netherlands
Nuclear	Phase-out	Minimal
Renewables	Growing	Moderate
Fossil Fuels	Higher	Higher
Per Capita CO ₂	Higher	Higher

Policy and energy mix choices — not just prosperity — determine per capita outcomes. France's heavy nuclear investment and the UK's growing renewables share keep their per capita emissions well below countries with similar GDP per capita.

Key Takeaways

1 Global CO₂ emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

2 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt). India (~3 Bt) is third and rising fast.

3 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.

4 Per capita inequalities remain extreme — a 150× gap separates the highest emitters from the lowest.

5 Policy and energy mix choices — not just prosperity — determine per capita outcomes.

Emissions are still rising, but the map of responsibility is shifting.